

Assignment Documentation

Factory Inventory & Production Management System (MVP)
Project manager

This document covers the project-manager assignment requirements: a lessons learned report and a practical functional artefact demonstration guide for a five-minute video.

1. Lessons Learned Report

1.1 Planning and Scope Control

- The MVP benefited from a clear scope boundary: authentication, inventory flows, production, dispatch, reports, and deployment readiness. Deferment of finance and advanced analytics prevented late-stage dilution.
- Feature ownership became more effective once role-based documentation and contribution mapping were aligned with actual code areas.

1.2 Technical Coordination

- Deployment behavior needed the same level of validation as feature behavior. Sprint 4 showed that a working local flow can still fail in production because of CORS and route mapping.
- Authentication and role-mapping needed coordination between Firebase setup, Mongo user records, protected client routes, and backend middleware; partial setup caused blocked access even when login succeeded.

1.3 Testing and Verification

- A stable verification set proved valuable: server build, client build, server tests, client tests, and repo-wide lint.
- Smoke testing the real business path—GRN confirmation, batch completion, order confirmation, dispatch FEFO, PDF download, and reports—was more meaningful than isolated UI checks.

1.4 Documentation and Handover

- Deployment and troubleshooting documents had to stay synchronized with the deployed architecture, especially after moving to Vercel serverless routing.
- Role-based sprint evidence improved traceability and made the final submission easier to defend during review.

1.5 Team Workflow Lessons

- Blocker-first communication was effective late in the project: production API issues were resolved faster when triaged separately from UI polish.
- The repository became easier to manage once changes were kept minimal and each role documented a clear responsibility boundary.

2. Functional Artefact Demonstration Guide (Maximum 5 Minutes)

2.1 Demonstration Objective

The objective of the video is to prove that the MVP is working as a role-based factory inventory and production management system, not to explain every internal detail. The demo should show one clean end-to-end flow and confirm deployment readiness.

2.2 Recommended Recording Tools

- Windows Xbox Game Bar for quick recording with minimal setup.
- OBS Studio for more control over capture area and microphone input.
- Clipchamp or CapCut only if trimming or adding captions is necessary after recording.

2.3 Manual Recording Setup

- Open the deployed client in one browser tab and keep the deployed API health endpoint available in another tab if needed.
- Prepare a seeded owner account and, if available, a wholesaler account so authentication and order visibility can be shown without delays.
- Use a browser zoom level that keeps tables readable and close unnecessary tabs or notifications before recording.

2.4 Exact 5-Minute Demo Sequence

Time	Screen / Action	What to Say
0:00 - 0:30	Open login page and sign in as owner.	Introduce the system as a TypeScript monorepo with Firebase authentication and role-based access.
0:30 - 1:00	Show the main navigation and briefly open Products.	Explain that the application covers products, suppliers, wholesalers, GRN, BOM, batches, orders, dispatch, reports, users, and audit.
1:00 - 1:40	Open GRNs and show a confirmed or ready-to-confirm goods received note.	State that raw-material stock enters the system through GRN confirmation and this action updates stock and audit history.
1:40 - 2:20	Open Batches and show create/start/complete flow or an existing completed batch.	Explain that production consumes raw materials using BOM-based consumption and creates finished-goods stock with batch traceability and expiry handling.
2:20 - 3:00	Open Orders and show wholesaler order records/status.	Describe that wholesalers can place orders while internal users confirm or cancel them, keeping order status separate from dispatch execution.
3:00 - 3:40	Open Dispatches and view a dispatch / delivery-note PDF action.	Explain FEFO allocation and note that the system can generate a delivery note PDF.
3:40 - 4:25	Open Reports and show stock, movements, expiry, sales-by-wholesaler, and traceability tabs.	State that reporting supports operational decisions, expiry monitoring, sales visibility, and wholesaler-to-batch traceability.
4:25 - 4:50	Open Users or Audit page.	Briefly show owner-only administration and audit visibility, emphasizing server-side role enforcement and action traceability.
4:50 - 5:00	Return to main layout and close.	Conclude that the MVP covers the main factory workflow from stock-in to production to dispatch with deployment-ready documentation.

2.5 Speaking Guide

- Keep narration factual and tied to what is visible on the screen.
- Do not spend time reading table rows; explain what each module proves about the business workflow.
- If a page contains seeded demo data, state that it was created using the documented seed process.

2.6 If the Video Cannot Be Recorded Immediately

If screen recording is not available in the environment, the team should still use the sequence above as a rehearsal script. The same timing, screens, and narration can be reproduced manually later using Xbox Game Bar or OBS Studio.

3. Conclusion

The project-manager assignment documentation is satisfied through a lessons learned report grounded in the real delivery history and a precise guide for recording a concise five-minute functional demonstration of the artefact.